

External Neumann problem for the Helmholtz equation

MG3416–Advanced Structural Acoustics - Lecture #10

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Navier-Stokes equations

- The mass, momentum, and energy conservation equations for a fluid flow (ignoring input mass, momentum and heat) read:

$$\left\{ \begin{array}{l} \frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v} , \\ \rho \frac{d\mathbf{v}}{dt} = \nabla \cdot \boldsymbol{\sigma} , \\ \rho T \frac{ds}{dt} = \boldsymbol{\tau} : \nabla \mathbf{v} - \nabla \cdot \mathbf{q} , \end{array} \right.$$

where $\frac{d}{dt} \equiv \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla$ and:

- $\mathbf{v}$ is the fluid velocity, ρ the density, T the temperature, s the (specific) entropy;
- $\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}$ is the stress tensor, p is the (static) fluid pressure, $\boldsymbol{\tau}$ is the viscous stress tensor;
- $\mathbf{q}$ is the heat flux vector.

Euler equations

- **Ideal fluid:** $\boldsymbol{\tau} = \mathbf{0}$, $\mathbf{q} = \mathbf{0}$, and the conservation equations then read:

$$\begin{cases} \frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v}, \\ \frac{d\mathbf{v}}{dt} = -\frac{1}{\rho} \nabla p, \\ \frac{ds}{dt} = 0. \end{cases}$$

- The flow is **isentropic** (each fluid particle has constant entropy), and by the equation of state $p = p(\rho, s)$:

$$\frac{dp}{dt} = c^2 \frac{d\rho}{dt}, \quad c^2(\rho, s) = \left. \frac{\partial p}{\partial \rho} \right|_s.$$

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Linearized Euler equations

- Linearization about a stationary fluid flow $(\varrho_0, \mathbf{v}_0, p_0)$:

$$\begin{cases} (\mathbf{v}_0 \cdot \nabla) \varrho_0 = -\varrho_0 \nabla \cdot \mathbf{v}_0, \\ (\mathbf{v}_0 \cdot \nabla) \mathbf{v}_0 = -\frac{1}{\varrho_0} \nabla p_0, \\ (\mathbf{v}_0 \cdot \nabla) p_0 = c_0^2 (\mathbf{v}_0 \cdot \nabla) \varrho_0. \end{cases}$$

- The actual flow is a perturbation $(\varrho', \mathbf{v}', p')$ of the stationary flow:

$$\begin{cases} \varrho = \varrho_0 + \varrho', \\ \mathbf{v} = \mathbf{v}_0 + \mathbf{v}', \\ p = p_0 + p'. \end{cases}$$

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Linearized Euler equations

- Let the stationary flow be at rest $\mathbf{v}_0 = \mathbf{0}$ in the absence of perturbations (waves), the **linearized Euler equations** finally yields:

$$\begin{cases} \frac{\partial \varrho'}{\partial t} = -\varrho_0 \nabla \cdot \mathbf{v}' \\ \frac{\partial \mathbf{v}'}{\partial t} = -\frac{1}{\varrho_0} \nabla p', \end{cases}$$

with $p' = c_0^2 \varrho'$ from the equation of state, taking $s(t) = s(0) = 0$ for each fluid particle.

- The operators $\frac{\partial}{\partial t}$ and $\nabla \cdot$ commute (but $\frac{d}{dt}$ and $\nabla \cdot$ do not in general), hence the **acoustic wave equation** reads:

$$\boxed{\frac{\partial}{\partial t} \left(\frac{1}{c_0^2} \frac{\partial p'}{\partial t} \right) - \varrho_0 \nabla \cdot \left(\frac{1}{\varrho_0} \nabla p' \right) = 0}.$$

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Notations and setting

■ Consider:

- an **acoustic fluid**: homogeneous, compressible, inviscid, gravity effects are neglected;
- irrotational motion $\nabla \times \mathbf{v}' = \mathbf{0}$, s.t. the fluid velocity $\mathbf{v}'$ reads $\mathbf{v}' = \nabla\psi$, thus the fluid pressure is $p' = -\varrho_0 \partial_t \psi$.

■ The **velocity potential** ψ is sought for as the solution of:

$$\left\{ \begin{array}{ll} \frac{1}{c_0^2} \frac{\partial^2 \psi}{\partial t^2} - \Delta \psi = g_{\text{IN}} & \text{in } \Omega_e, \\ \frac{\partial \psi}{\partial \mathbf{n}} = v & \text{on } \partial\Omega_e = \Gamma, \\ \psi(\cdot, 0) = 0 & \text{in } \Omega_e, \\ \dot{\psi}(\cdot, 0) = 0 & \text{in } \Omega_e, \end{array} \right.$$

with c_0 : sound speed of the fluid at rest.

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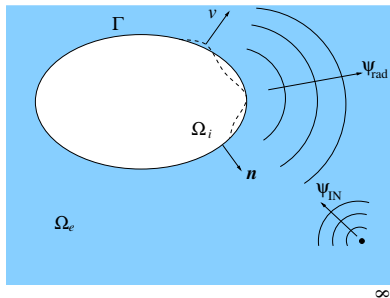
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- The velocity potential reads:

$$\psi(\mathbf{x}, t) = \psi_{IN}(\mathbf{x}, t) + \psi_{d0}(\mathbf{x}, t) + \psi_{rad}(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega_e \times \mathbb{R},$$

where:

- ψ_{IN} : incident potential (data);
- ψ_{d0} : diffracted potential, the structure being motionless;
- ψ_{rad} : radiated potential.

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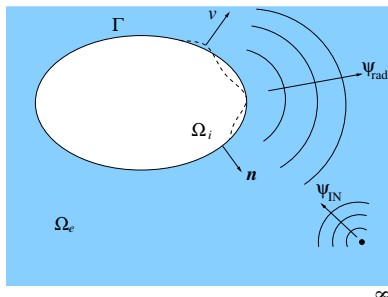
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- The diffracted potential ψ_{d0} satisfies:

$$\begin{cases} \frac{1}{c_0^2} \frac{\partial^2 \psi_{d0}}{\partial t^2} - \Delta \psi_{d0} = 0 & \text{in } \Omega_e, \\ \frac{\partial \psi_{d0}}{\partial \mathbf{n}} = -\frac{\partial \psi_{IN}}{\partial \mathbf{n}} & \text{on } \Gamma. \end{cases}$$

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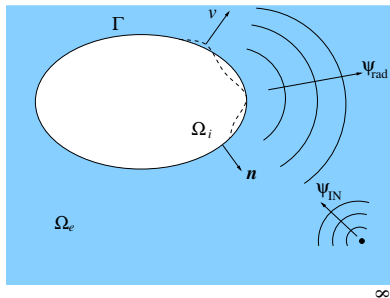
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- The radiated potential ψ_{rad} satisfies:

$$\left\{ \begin{array}{l} \frac{1}{c_0^2} \frac{\partial^2 \psi_{\text{rad}}}{\partial t^2} - \Delta \psi_{\text{rad}} = 0 \quad \text{in } \Omega_e, \\ \frac{\partial \psi_{\text{rad}}}{\partial \mathbf{n}} = v \quad \text{on } \Gamma. \end{array} \right.$$

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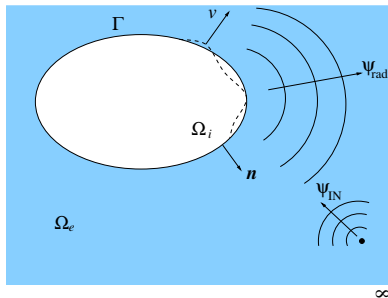
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- The **incident potential** ψ_{IN} satisfies:

$$\frac{1}{c_0^2} \frac{\partial^2 \psi_{\text{IN}}}{\partial t^2} - \Delta \psi_{\text{IN}} = g_{\text{IN}}(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \mathbb{R}^3 \times \mathbb{R},$$

for some sound source g_{IN} **in the full physical space.**

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Frequency domain formulation

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- The Fourier transform:

$$\hat{\psi}(\mathbf{x}, \omega) = \int_{\mathbb{R}} e^{-i\omega t} \psi(\mathbf{x}, t) dt, \quad \psi(\mathbf{x}, t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega t} \hat{\psi}(\mathbf{x}, \omega) d\omega.$$

- In the frequency domain, for $k = \frac{\omega}{c_0}$:

$$\left\{ \begin{array}{ll} \Delta \hat{\psi} + k^2 \hat{\psi} = 0 & \text{in } \Omega_e, \\ \frac{\partial \hat{\psi}}{\partial \mathbf{n}} = u & \text{on } \Gamma, \\ \left| \hat{\psi} \right| = O\left(\frac{1}{r}\right), \quad \left| \frac{\partial \hat{\psi}}{\partial r} + ik\hat{\psi} \right| = O\left(\frac{1}{r^2}\right) & \text{as } r = \|\mathbf{x}\| \rightarrow +\infty. \end{array} \right.$$

- Sommerfeld radiation conditions "at infinity": the radiated waves are almost plane and do not back propagate toward Γ .

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- The external Helmholtz problem admits a **unique** solution $\forall k \in \mathbb{R}$, hence:

- There exists a linear operator $\mathcal{B}_\Gamma(k) : \mathcal{C}'_\Gamma \rightarrow \mathcal{C}_\Gamma$ s.t.:

$$\hat{\psi}|_\Gamma = \mathcal{B}_\Gamma(k)u, \quad \text{on } \Gamma;$$

- There exists a linear operator $\mathcal{R}_x(k) : \mathcal{C}'_\Gamma \rightarrow \mathbb{C}$ s.t.:

$$\hat{\psi}(x) = \mathcal{R}_x(k)u, \quad x \in \Omega_e,$$

with $\mathcal{C}_\Gamma$: the set of admissible fields on Γ ($\mathcal{C}'_\Gamma$: its dual¹).

- $\mathcal{Z}_\Gamma(\omega) = -i\omega \varrho_0 \mathcal{B}_\Gamma(\frac{\omega}{c_0})$: **boundary impedance operator**.
- $\mathcal{Z}_x(\omega) = -i\omega \varrho_0 \mathcal{R}_x(\frac{\omega}{c_0})$: **radiation impedance operator**.

¹ $\mathcal{C}_\Gamma = H^{1/2}(\Gamma)$ the set of traces in $H^1_{\text{loc}}(\Omega_e)$, $\mathcal{C}'_\Gamma = H^{-1/2}(\Gamma)$ s.t. $\mathcal{C}_\Gamma \subset \mathbb{L}^2(\Gamma) \subset \mathcal{C}'_\Gamma$

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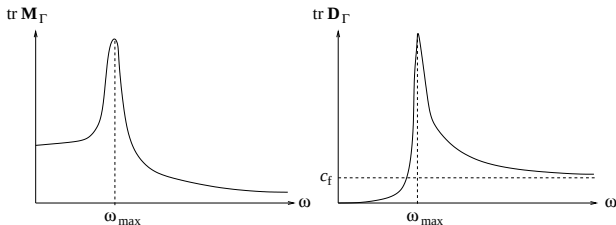
Boundary impedance

- The boundary impedance $\mathbf{Z}_\Gamma(\omega)$ is symmetric and reads:

$$i\omega \mathbf{Z}_\Gamma(\omega) = -\omega^2 \mathbf{M}_\Gamma \left(\frac{\omega}{c_0} \right) + i\omega \mathbf{D}_\Gamma \left(\frac{\omega}{c_0} \right),$$

where:

- The **reactive part** $\omega \mapsto \text{Tr } \mathbf{M}_\Gamma \left(\frac{\omega}{c_0} \right)$ (left) is generally unsigned, though it is positive if $\mathbb{R}^3 \setminus \overline{\Omega}_e$ is convex;
- The **resistive part** $\omega \mapsto \text{Tr } \mathbf{D}_\Gamma \left(\frac{\omega}{c_0} \right)$ (right) is positive.



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- Diffracted velocity potential:

$$\begin{aligned}\hat{\psi}_{\text{d0}}(\omega)|_{\Gamma} &= -\mathcal{B}_{\Gamma} \left(\frac{\omega}{c_0} \right) \frac{\partial \hat{\psi}_{\text{IN}}}{\partial \mathbf{n}}, \quad \mathbf{x} \in \Gamma, \\ \hat{\psi}_{\text{d0}}(\mathbf{x}, \omega) &= -\mathcal{R}_{\mathbf{x}} \left(\frac{\omega}{c_0} \right) \frac{\partial \hat{\psi}_{\text{IN}}}{\partial \mathbf{n}}, \quad \mathbf{x} \in \Omega_e.\end{aligned}$$

- Radiated velocity potential:

$$\begin{aligned}\hat{\psi}_{\text{rad}}(\omega)|_{\Gamma} &= \mathcal{B}_{\Gamma} \left(\frac{\omega}{c_0} \right) \hat{v}(\omega), \quad \mathbf{x} \in \Gamma, \\ \hat{\psi}_{\text{rad}}(\mathbf{x}, \omega) &= \mathcal{R}_{\mathbf{x}} \left(\frac{\omega}{c_0} \right) \hat{v}(\omega), \quad \mathbf{x} \in \Omega_e.\end{aligned}$$

- (Linear) **scattering operator** $\mathcal{T}_{\Gamma}(\omega) \equiv \mathbf{I} - \mathcal{B}_{\Gamma}(\frac{\omega}{c_0})\partial_{\mathbf{n}}$ s.t.:

$$(\hat{\psi}_{\text{IN}}(\omega) + \hat{\psi}_{\text{d0}}(\omega))|_{\Gamma} = \mathcal{T}_{\Gamma}(\omega)\hat{\psi}_{\text{IN}}(\omega)|_{\Gamma}.$$

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Acoustic Green's function

- The **Green's function** $\mathcal{G}(\mathbf{x}, \mathbf{y})$ is the solution of:

$$\left\{ \begin{array}{ll} \Delta_{\mathbf{x}} \mathcal{G} + k^2 \mathcal{G} = -\delta_{\mathbf{y}}(\mathbf{x}) & \text{in } \mathbb{R}^3, \\ |\mathcal{G}| = O\left(\frac{1}{r}\right), \quad \left| \frac{\partial \mathcal{G}}{\partial r} + ik\mathcal{G} \right| = O\left(\frac{1}{r^2}\right) & \text{as } r = \|\mathbf{x}\| \rightarrow +\infty, \end{array} \right.$$

for some $\mathbf{y} \in \mathbb{R}^3$ fixed.

- The **unique** outward solution (which do not back propagate toward the source at $\mathbf{y} \in \mathbb{R}^3$) is:

$$\mathcal{G}(\mathbf{x}, \mathbf{y}) = G(\mathbf{x} - \mathbf{y}) = \frac{e^{-ik\|\mathbf{x} - \mathbf{y}\|}}{4\pi\|\mathbf{x} - \mathbf{y}\|}.$$

- **Remarks:** $\mathcal{G}$ is singular at $\mathbf{x} = \mathbf{y}$. In an inhomogeneous medium $\mathcal{G}(\mathbf{x}, \mathbf{y}) \neq G(\mathbf{x} - \mathbf{y})$ in general.

Single and double layer potentials

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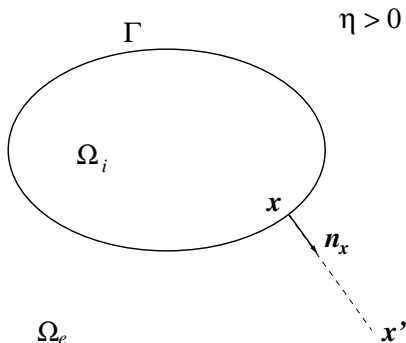
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- For $x' \in \mathbb{R}^3$, let $x' = x + \eta n_x$ where $x \in \Gamma$ and n_x is the outward unit normal to Γ at x ; hence:
 - $x' \in \Omega_e$ if $\eta > 0$ (see figure),
 - $x' \in \Omega_i$ if $\eta < 0$.

Single and double layer potentials

Definitions

- Let $\psi : \Gamma \rightarrow \mathbb{C}$ continuous; the **single layer potential** is:

$$\psi_S(\mathbf{x}') = \int_{\Gamma} G(\mathbf{x}' - \mathbf{y}) \psi(\mathbf{y}) dS(\mathbf{y}),$$

and $\mathbf{x}' \mapsto \psi_S(\mathbf{x}')$ is continuous in $\mathbb{R}^3$.

- Let $\psi : \Gamma \rightarrow \mathbb{C}$ continuous; the **double layer potential** is:

$$\psi_D(\mathbf{x}') = \int_{\Gamma} \frac{\partial G(\mathbf{x}' - \mathbf{y})}{\partial \mathbf{n}_{\mathbf{y}}} \psi(\mathbf{y}) dS(\mathbf{y}).$$

and $\mathbf{x}' \mapsto \psi_D(\mathbf{x}')$ is continuous in $\mathbb{R}^3 \setminus \Gamma$.

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Single and double layer operators

Definitions

- The **single layer operator** $\mathcal{S}_S(k) : \mathcal{C}'_\Gamma \rightarrow \mathcal{C}_\Gamma$:

$$\mathcal{S}_S(k)\psi(\mathbf{x}) = \int_\Gamma G(\mathbf{x} - \mathbf{y}; k)\psi(\mathbf{y})dS(\mathbf{y}).$$

- The **double layer operator** $\mathcal{S}_D(k) : \mathcal{C}_\Gamma \rightarrow \mathcal{C}_\Gamma$:

$$\mathcal{S}_D(k)\psi(\mathbf{x}) = \int_\Gamma \frac{\partial G(\mathbf{x} - \mathbf{y}; k)}{\partial \mathbf{n}_y} \psi(\mathbf{y})dS(\mathbf{y}).$$

- The normal derivative of $\mathcal{S}_D(k)$, $\mathcal{S}_T(k) : \mathcal{C}_\Gamma \rightarrow \mathcal{C}'_\Gamma$:

$$\mathcal{S}_T(k)\psi(\mathbf{x}) = - \oint_\Gamma \frac{\partial^2 G(\mathbf{x} - \mathbf{y}; k)}{\partial \mathbf{n}_x \partial \mathbf{n}_y} \psi(\mathbf{y})dS(\mathbf{y}).$$

- **Remark:** $G(\mathbf{x} - \mathbf{y}; k) = G(\mathbf{y} - \mathbf{x}; k)$, thus $\mathcal{S}_S(k)$ and $\mathcal{S}_T(k)$ are **symmetric**.

Single layer potential

Properties of the traces

- Let $\eta > 0$ i.e. $\mathbf{x}' \in \Omega_e$. Then:

$$\lim_{\eta \rightarrow 0^+} \psi_S(\mathbf{x}') := \psi_S^+(\mathbf{x}) = \mathbf{S}_S(k)\psi(\mathbf{x}),$$

$$\lim_{\eta \rightarrow 0^+} \frac{\partial \psi_S(\mathbf{x}')}{\partial \mathbf{n}_x} := \frac{\partial \psi_S^+}{\partial \mathbf{n}_x}(\mathbf{x}) = \left(\frac{1}{2} \mathbf{I} + \mathbf{S}_D^\top(k) \right) \psi(\mathbf{x}).$$

- Let $\eta < 0$ i.e. $\mathbf{x}' \in \Omega_i$. Then:

$$\lim_{\eta \rightarrow 0^-} \psi_S(\mathbf{x}') := \psi_S^-(\mathbf{x}) = \mathbf{S}_S(k)\psi(\mathbf{x}),$$

$$\lim_{\eta \rightarrow 0^-} \frac{\partial \psi_S(\mathbf{x}')}{\partial \mathbf{n}_x} := \frac{\partial \psi_S^-}{\partial \mathbf{n}_x}(\mathbf{x}) = \left(-\frac{1}{2} \mathbf{I} + \mathbf{S}_D^\top(k) \right) \psi(\mathbf{x}).$$

- Define the **jump** $[[\psi]] = \psi^+ - \psi^-$; hence:

$$[[\psi_S]] = 0, \quad \left[\left[\frac{\partial \psi_S}{\partial \mathbf{n}_x} \right] \right] = \psi.$$

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Double layer potential

Properties of the traces

- Let $\eta > 0$ i.e. $\mathbf{x}' \in \Omega_e$. Then:

$$\lim_{\eta \rightarrow 0^+} \psi_D(\mathbf{x}') := \psi_D^+(\mathbf{x}) = \left(-\frac{1}{2}\mathbf{I} + \mathbf{S}_D(k) \right) \psi(\mathbf{x}),$$

$$\lim_{\eta \rightarrow 0^+} \frac{\partial \psi_D(\mathbf{x}')}{\partial \mathbf{n}_x} := \frac{\partial \psi_D^+}{\partial \mathbf{n}_x}(\mathbf{x}) = -\mathbf{S}_T(k) \psi(\mathbf{x}).$$

- Let $\eta < 0$ i.e. $\mathbf{x}' \in \Omega_i$. Then:

$$\lim_{\eta \rightarrow 0^-} \psi_D(\mathbf{x}') := \psi_D^-(\mathbf{x}) = \left(\frac{1}{2}\mathbf{I} + \mathbf{S}_D(k) \right) \psi(\mathbf{x}),$$

$$\lim_{\eta \rightarrow 0^-} \frac{\partial \psi_D(\mathbf{x}')}{\partial \mathbf{n}_x} := \frac{\partial \psi_D^-}{\partial \mathbf{n}_x}(\mathbf{x}) = -\mathbf{S}_T(k) \psi(\mathbf{x}).$$

- Define the **jump** $[[\psi]] = \psi^+ - \psi^-$; hence:

$$[[\psi_D]] = -\psi, \quad \left[\left[\frac{\partial \psi_D}{\partial \mathbf{n}_x} \right] \right] = 0.$$

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Outer Helmholtz equation

Boundary integral equations

- Let ψ_+ be a solution of the Helmholtz equation in Ω_e with Sommerfeld radiation conditions:

$$\left\{ \begin{array}{ll} \Delta \psi_+ + k^2 \psi_+ = 0 & \text{in } \Omega_e, \\ |\psi_+| = O\left(\frac{1}{r}\right), \quad \left| \frac{\partial \psi_+}{\partial r} + ik\psi_+ \right| = O\left(\frac{1}{r^2}\right) & \text{as } r = \|\mathbf{x}\| \rightarrow +\infty. \end{array} \right.$$

- Then the following integral equations hold for $\mathbf{x} \in \mathbb{R}^3$:

$$\varepsilon_+ \psi_+(\mathbf{x}) = \int_{\Gamma} \left(G(\mathbf{x} - \mathbf{y}) \frac{\partial \psi_+}{\partial \mathbf{n}_y}(\mathbf{y}) - \frac{\partial G(\mathbf{x} - \mathbf{y})}{\partial \mathbf{n}_y} \psi_+(\mathbf{y}) \right) dS(\mathbf{y}),$$

where—provided that $\Gamma = \partial\Omega_e$ is regular enough:

$$\varepsilon_+ = 1 \quad \text{if } \mathbf{x} \in \Omega_e,$$

$$\varepsilon_+ = \frac{1}{2} \quad \text{if } \mathbf{x} \in \Gamma,$$

$$\varepsilon_+ = 0 \quad \text{if } \mathbf{x} \in \Omega_i.$$

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Boundary integral equations

- From the foregoing **Boundary Integral Equation** (BIE for $\boldsymbol{x} \in \Gamma$), the solution of the external Helmholtz problem (s.t. $\frac{\partial \psi}{\partial \boldsymbol{n}} = u$ on Γ) should satisfy:

$$\left(\frac{1}{2} \boldsymbol{I} + \boldsymbol{S}_D(k) \right) \hat{\psi} = \boldsymbol{S}_S(k) u \quad \text{in } \mathcal{C}_\Gamma. \quad [\text{A}]$$

- Owing to the trace properties of the single and double layer potentials, it should also satisfy:

$$\boldsymbol{S}_T(k) \hat{\psi} = \left(\frac{1}{2} \boldsymbol{I} - \boldsymbol{S}_D^\top(k) \right) u \quad \text{in } \mathcal{C}'_\Gamma. \quad [\text{B}]$$

- **Remark:** it is not guaranteed that the left-hand side operators are invertible.

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Inner Helmholtz equation

Boundary integral equations

- Let ψ_- be a solution of the Helmholtz equation in Ω_i :

$$\Delta\psi_- + k^2\psi_- = 0 \quad \text{in } \Omega_i.$$

- Then the following integral equations hold for $\mathbf{x} \in \mathbb{R}^3$:

$$\varepsilon_- \psi_-(\mathbf{x}) = - \int_{\Gamma} \left(G(\mathbf{x} - \mathbf{y}) \frac{\partial \psi_-}{\partial \mathbf{n}_y}(\mathbf{y}) - \frac{\partial G(\mathbf{x} - \mathbf{y})}{\partial \mathbf{n}_y} \psi_-(\mathbf{y}) \right) dS(\mathbf{y}),$$

where—provided that $\Gamma = \partial\Omega_i$ is regular enough:

$$\varepsilon_- = 1 \quad \text{if } \mathbf{x} \in \Omega_i,$$

$$\varepsilon_- = \frac{1}{2} \quad \text{if } \mathbf{x} \in \Gamma,$$

$$\varepsilon_- = 0 \quad \text{if } \mathbf{x} \in \Omega_e.$$

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- From the foregoing [Boundary Integral Equation](#) (BIE for $\boldsymbol{x} \in \Gamma$), the solution of the inner Helmholtz equation is given as:

$$\left(\frac{1}{2}\boldsymbol{I} - \boldsymbol{S}_D(k)\right)\psi_- = -\boldsymbol{S}_S(k)\frac{\partial\psi_-}{\partial\boldsymbol{n}} \quad \text{in } \mathcal{C}_\Gamma. \quad [\text{C}]$$

- Owing to the trace properties of the single and double layer potentials, it is also the solution of:

$$\boldsymbol{S}_T(k)\psi_- = -\left(\frac{1}{2}\boldsymbol{I} + \boldsymbol{S}_D^\top(k)\right)\frac{\partial\psi_-}{\partial\boldsymbol{n}} \quad \text{in } \mathcal{C}'_\Gamma. \quad [\text{D}]$$

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Internal Neumann problem

Frequency domain formulation

- In the frequency domain, for $k = \frac{\omega}{c_0}$:

$$\left\{ \begin{array}{ll} \Delta \psi_- + k^2 \psi_- = 0 & \text{in } \Omega_i, \\ \frac{\partial \psi_-}{\partial \mathbf{n}} = 0 & \text{on } \Gamma, \\ \int_{\Omega_i} \psi_- \, d\mathbf{x} = 0 \end{array} \right.$$

- **Remark:** the last condition evades constant solutions. Indeed, the weak form of the Helmholtz equation reads

$$\int_{\Omega_i} \nabla \psi_- \cdot \nabla \varphi \, d\mathbf{x} = k^2 \int_{\Omega_i} \psi_- \varphi \, d\mathbf{x}, \quad \forall \varphi \in V \subset H^1(\Omega_i, \mathbb{R}),$$

so that constant solutions associated to the eigenvalue 0 are excluded by this condition.

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Spectral theorem

- **Spectral theorem:** there exists a countable set of eigenvalues $0 < \lambda_1 < \lambda_2 < \dots < \lambda_\alpha \dots$ ($\lambda_\alpha = k_\alpha^2$) each of multiplicity n_α , and associated eigenfunctions $\psi_\alpha^1, \psi_\alpha^2, \dots, \psi_\alpha^{n_\alpha}$ in:

$$V = \left\{ \psi \in H^1(\Omega_i, \mathbb{R}); \int_{\Omega_i} \psi d\mathbf{x} = 0 \right\}$$

such that:

$$\begin{cases} -\Delta \psi_\alpha^j = \lambda_\alpha \psi_\alpha^j & \text{in } \Omega_i, \\ \frac{\partial \psi_\alpha^j}{\partial \mathbf{n}} = 0 & \text{on } \Gamma, \end{cases}$$

$$\forall j \in \{1, 2, \dots, n_\alpha\}.$$

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- From the BIEs [C] and [D] of the inner Helmholtz problem it is deduced that:

$$\ker \mathbf{S}_T(k_\alpha) = \ker \left(\frac{1}{2} \mathbf{I} - \mathbf{S}_D(k_\alpha) \right) = \mathcal{C}_\alpha \subset \mathcal{C}_\Gamma,$$

where $\mathcal{C}_\alpha = \text{span}\{\psi_\alpha^1|_\Gamma, \psi_\alpha^2|_\Gamma, \dots, \psi_\alpha^{n_\alpha}|_\Gamma\}$, with $\#\mathcal{C}_\alpha = n_\alpha$.

- Remind the BIE [B] $\mathbf{S}_T(k)\tilde{\psi} = (\frac{1}{2}\mathbf{I} - \mathbf{S}_D^\top(k))u$, then:
 - If $k \neq k_\alpha$ (for all α): $\ker \mathbf{S}_T(k) = \{\mathbf{0}\}$ and [B] is uniquely invertible;
 - If $k = k_\alpha$ (for some α): [B] is invertible iff the right-hand side is orthogonal to $\mathcal{C}_\alpha$, which is actually true since:

$$\left\langle \left(\frac{1}{2} \mathbf{I} - \mathbf{S}_D^\top(k_\alpha) \right) u, \psi_\alpha^j \right\rangle_\Gamma = \left\langle u, \left(\frac{1}{2} \mathbf{I} - \mathbf{S}_D(k_\alpha) \right) \psi_\alpha^j \right\rangle_\Gamma = 0.$$

Invertible BIE for the external Neumann problem

- Consequently, the solutions of the BIE [B] read:

$$\begin{aligned}\tilde{\psi}(\omega) &= \hat{\psi}(\omega)|_{\Gamma} && \text{if } \omega \neq \omega_{\alpha}, \\ \tilde{\psi}(\omega) &= \hat{\psi}(\omega_{\alpha})|_{\Gamma} + \sum_{j=1}^{n_{\alpha}} C_j \psi_{\alpha}^j|_{\Gamma} && \text{if } \omega = \omega_{\alpha},\end{aligned}$$

where $C_j \in \mathbb{C}$ for $1 \leq j \leq n_{\alpha}$ and $k_{\alpha} = \frac{\omega_{\alpha}}{c_0}$.

- A symmetric BIE for the solution $\hat{\psi}$ of the external Neumann problem that is invertible **for all frequencies** is given by (Angélini-Hutin 1983):

$$\begin{pmatrix} 0 \\ \hat{\psi}|_{\Gamma} \end{pmatrix} = \begin{bmatrix} -\mathbf{S}_T(\frac{\omega}{c_0}) & \frac{1}{2}\mathbf{I} - \mathbf{S}_D^T(\frac{\omega}{c_0}) \\ \frac{1}{2}\mathbf{I} - \mathbf{S}_D(\frac{\omega}{c_0}) & \mathbf{S}_S(\frac{\omega}{c_0}) \end{bmatrix} \begin{pmatrix} \tilde{\psi} \\ u \end{pmatrix}.$$

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- 1 If $k \neq k_\alpha$ then the first line has a unique solution $\tilde{\psi} = \hat{\psi}|_\Gamma$ and the second line reads:

$$\begin{aligned}\left(\frac{1}{2}\mathbf{I} - \mathbf{S}_D(k)\right)\tilde{\psi} + \mathbf{S}_S(k)u &= \left(\frac{1}{2}\mathbf{I} - \mathbf{S}_D(k)\right)\hat{\psi} + \mathbf{S}_S(k)u \\ &= \hat{\psi}|_\Gamma - \left(\frac{1}{2}\mathbf{I} + \mathbf{S}_D(k)\right)\hat{\psi} + \mathbf{S}_S(k)u \\ &= \hat{\psi}|_\Gamma,\end{aligned}$$

owing to the BIE [A].

- 2 If $k = k_\alpha$ then $\tilde{\psi}(\omega_\alpha) = \hat{\psi}(\omega_\alpha)|_\Gamma + \sum_{j=1}^{n_\alpha} C_j \psi_\alpha^j|_\Gamma$, thus:

$$\left(\frac{1}{2}\mathbf{I} - \mathbf{S}_D(k_\alpha)\right)\tilde{\psi} + \mathbf{S}_S(k_\alpha)u = \left(\frac{1}{2}\mathbf{I} - \mathbf{S}_D(k_\alpha)\right)\hat{\psi} + \mathbf{S}_S(k_\alpha)u,$$

and the same conclusion holds.

Construction of the impedance operators

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- Solving the symmetric BIE system by eliminating $\tilde{\psi}$ yields $\hat{\psi}|_{\Gamma}$ in terms of u , hence the boundary impedance operator $\mathcal{B}_{\Gamma}(k)$ (or $\mathcal{B}_{\Gamma}(k_{\alpha})$).
- Consider the RIE for the outer Helmholtz problem applied to $\hat{\psi}$, the solution of the external Neumann problem:

$$\begin{aligned}\hat{\psi}(\mathbf{x}) &= \int_{\Gamma} G(\mathbf{x} - \mathbf{y}, k) u(\mathbf{y}) dS(\mathbf{y}) - \int_{\Gamma} \frac{\partial G(\mathbf{x} - \mathbf{y}, k)}{\partial \mathbf{n}_{\mathbf{y}}} \hat{\psi}(\mathbf{y}) dS(\mathbf{y}), \quad \mathbf{x} \in \Omega_e \\ &= \mathcal{R}_{\mathbf{x}}^1(k) u - \mathcal{R}_{\mathbf{x}}^2(k) \hat{\psi}|_{\Gamma} \\ &= \mathcal{R}_{\mathbf{x}}^1(k) u - \mathcal{R}_{\mathbf{x}}^2(k) \mathcal{B}_{\Gamma}(k) u,\end{aligned}$$

with the definitions $\mathcal{R}_{\mathbf{x}}^1(k) u = \int_{\Gamma} G(\mathbf{x} - \mathbf{y}, k) u(\mathbf{y}) dS(\mathbf{y})$
and $\mathcal{R}_{\mathbf{x}}^2(k) \psi = \int_{\Gamma} \frac{\partial G(\mathbf{x} - \mathbf{y}, k)}{\partial \mathbf{n}_{\mathbf{y}}} \psi(\mathbf{y}) dS(\mathbf{y})$ for $\mathbf{x} \in \Omega_e$.

Hence:

$$\mathcal{R}_{\mathbf{x}}(k) = \mathcal{R}_{\mathbf{x}}^1(k) - \mathcal{R}_{\mathbf{x}}^2(k) \mathcal{B}_{\Gamma}(k).$$

Numerical issues

FE discretization

- Finite-dimensional discretization of the boundary operators in $\mathcal{C}_N = \text{span}\{\psi_1, \psi_2, \dots, \psi_N\}$ with $\#\mathcal{C}_N = N$:

$$\begin{aligned} [S(k)]_{mn} &= \langle \mathbf{S}_S(k) \psi_m, \psi_n \rangle_\Gamma \\ &= \iint_{\Gamma \times \Gamma} G(\mathbf{x} - \mathbf{y}, k) \psi_m(\mathbf{y}) \psi_n(\mathbf{x}) dS(\mathbf{x}) dS(\mathbf{y}), \\ [D(k)]_{mn} &= \langle \mathbf{S}_D(k) \psi_m, \psi_n \rangle_\Gamma \\ &= \iint_{\Gamma \times \Gamma} \frac{\partial G(\mathbf{x} - \mathbf{y}, k)}{\partial \mathbf{n}_y} \psi_m(\mathbf{y}) \psi_n(\mathbf{x}) dS(\mathbf{x}) dS(\mathbf{y}). \end{aligned}$$

- **Petrov-Galerkin method** whereby the trial and test spaces are different, *e.g.* **collocation approach** using test functions of the form $\psi_n(\mathbf{x}) = \delta(\mathbf{x} - \boldsymbol{\xi}_n)$.

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Regularization of the normal derivative of the double layer potential

- The normal derivative of the double layer potential $\mathbf{S}_T(k)$ is singular and is understood as its Cauchy principal value:

$$\mathbf{S}_T(k)\psi(\mathbf{x}) = - \lim_{\epsilon \rightarrow 0} \int_{\Gamma \setminus B(\mathbf{y}, \epsilon)} \frac{\partial^2 G(\mathbf{x} - \mathbf{y}, k)}{\partial \mathbf{n}_x \partial \mathbf{n}_y} \psi(\mathbf{y}) dS(\mathbf{y}),$$

where $B(\mathbf{y}, \epsilon)$ is the open ball of radius ϵ centered in $\mathbf{y}$.

- Regularization formula (Maue 1949, Mitzner 1966):

$$\begin{aligned} \langle \mathbf{S}_T(k)\psi, \phi \rangle_{\Gamma} &= -k^2 \iint_{\Gamma \times \Gamma} G(\mathbf{x} - \mathbf{y}, k) (\psi(\mathbf{y}) \mathbf{n}_y) \cdot (\phi(\mathbf{x}) \mathbf{n}_x) dS(\mathbf{x}) dS(\mathbf{y}) \\ &+ \iint_{\Gamma \times \Gamma} G(\mathbf{x} - \mathbf{y}, k) (\mathbf{n}_y \times \nabla_y \psi(\mathbf{y})) \cdot (\mathbf{n}_x \times \nabla_x \phi(\mathbf{x})) dS(\mathbf{x}) dS(\mathbf{y}). \end{aligned}$$

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- Discretized version of the symmetric BIE in the finite-dimensional subspace $\mathcal{C}_N \subset \mathcal{C}_\Gamma \subset \mathcal{C}'_\Gamma$:

$$\begin{pmatrix} 0 \\ \Psi_\Gamma \end{pmatrix} = \begin{bmatrix} -[T(\frac{\omega}{c_0})] & \frac{1}{2}[E]^\top - [D(\frac{\omega}{c_0})]^\top \\ \frac{1}{2}[E] - [D(\frac{\omega}{c_0})] & [S(\frac{\omega}{c_0})] \end{bmatrix} \begin{pmatrix} \tilde{\Psi} \\ U \end{pmatrix}.$$

- Gauss elimination of $\tilde{\Psi}$ in the $2N \times 2N$ system above:
 - If $\omega \neq \omega_\alpha$, $[T(\frac{\omega}{c_0})]$ is invertible and Gauss elimination is stopped at row N to yield $\Psi_\Gamma = [B_\Gamma(\frac{\omega}{c_0})]U$, where $[B_\Gamma(\frac{\omega}{c_0})]$ is the $N \times N$ complex symmetric matrix corresponding to the discretization of $\mathcal{B}_\Gamma(\frac{\omega}{c_0})$ in $\mathcal{C}_N$;
 - If $\omega = \omega_\alpha$, $[T(\frac{\omega}{c_0})]$ is not invertible and Gauss elimination is stopped at row $N - n_\alpha$ —practically when a null pivot arises because n_α is *a priori* unknown.

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